DIRECT SEMISTABLE A1-CONCORDANCE AND A1-CONNECTEDNESS OF FIXED-DETERMINANT VECTOR-BUNDLE MODULI STACKS

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Allowed standard prerequisites (editorial).

- A1 homotopy, algebraic vector-bundle stacks, semistability, extensions and concordance. Original complete author formulations, construction proofs and bibliography retained.

1. INTRODUCTION

Let C be an irreducible smooth projective curve of genus $g \geq 2$ over an algebraically closed field k. Let $\mathcal{M}_C(n, L)$ denote the moduli stack of vector bundles on C of rank n and determinant $L \in \mathrm{Pic}(C)$. By definition, an object in $\mathcal{M}_C(n, L)$ over a k-scheme S is given by a rank n vector bundle $\mathcal{E}$ on $C \times S$, together with an isomorphism $\det(\mathcal{E}) \xrightarrow{\simeq} p^*L$, where $p : C \times S \longrightarrow C$ is the first projection. We have the following inclusions of open substacks:

$$\mathcal{M}_C^s(n, L) \subset \mathcal{M}_C^{ss}(n, L) \subset \mathcal{M}_C(n, L),$$

where $\mathcal{M}_C^{ss}(n, L)$ (respectively, $\mathcal{M}_C^s(n, L)$) denotes the stack of semistable (respectively, stable) vector bundles of rank n and determinant L, whose objects over a k-scheme S are given by a vector bundle $\mathcal{E}$ on $C \times S$ whose restriction to each geometric point $\mathrm{Spec}\, K \to S$ is a *semistable* (respectively, stable) vector bundle over the curve C_K (see Definition 2.8 for details). It is known that $\mathcal{M}_C(n, L)$ is a smooth algebraic stack over k.

Via the nerve construction, any algebraic stack defines an object in the $\mathbb{A}^1$–homotopy category [MV] by regarding it as a simplicial sheaf. This naturally raises the question of $\mathbb{A}^1$–connectedness of the stacks $\mathcal{M}_C(n, L)$, as well as for the open substacks $\mathcal{M}_C^{ss}(n, L)$ and $\mathcal{M}_C^s(n, L)$. In [HY], the $\mathbb{A}^1$–connectedness of $\mathcal{M}_C(n, L)$ was proved. It was shown that two vector bundles of the same rank are $\mathbb{A}^1$–concordant (see Definition 2.5) if and only if their determinants are isomorphic. This was achieved through looking at extensions of vector bundles rather than individual vector bundles, and showing that, in a certain sense, any extension of vector bundles can be "joined" to the trivial extension via a line contained in the moduli space of extensions between two fixed vector bundles over a projective scheme. The key observation here was the fact that the moduli space of such extensions of vector bundles is an affine space, which allowed for such a result. Unfortunately, this approach does not carry through for the open substack $\mathcal{M}_C^{ss}(n, L)$ or $\mathcal{M}_C^s(n, L)$, as the presence of semistability causes additional complications. It is also worth mentioning, as pointed out in [HY, Example 2.9], that when n and $\deg(L)$ are coprime, the $\mathbb{A}^1$-connectedness of the stable moduli stack $\mathcal{M}_C^s(n, L)$ cannot be concluded directly from the rationality of its coarse moduli space alone.

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There is also the notion of (quasi-)parabolic vector bundles over a curve introduced by Mehta and Seshadri [MS]; these are vector bundles together with the data of weighted flags on the fibers over a collection of finitely many points on the curve. From the perspective of algebraic geometry, the study of the moduli stacks of (quasi-)parabolic vector bundles is an important and active area of research.

In this article, we aim to address the question of $\mathbb{A}^1$–connectedness of the following moduli stacks:

- The moduli stack $\mathcal{M}_C^{ss}(n, L)$ of semistable vector bundles over C of fixed rank n and determinant L.
- The moduli stack $\mathcal{PM}_C^{\boldsymbol{e},\boldsymbol{m}}(n, L)$ of quasi-parabolic vector bundles on C with a fixed determinant and a quasi-parabolic data $(\boldsymbol{e}, \boldsymbol{m})$ along a set of parabolic points on C (see Section 3 for details).
- The stack $\mathcal{PM}_C^{\boldsymbol{\alpha}-ss,\boldsymbol{e},\boldsymbol{m}}(n, L)$ of $\boldsymbol{\alpha}$-semistable parabolic vector bundles for a system of weights $\boldsymbol{\alpha}$; these are open substacks of $\mathcal{PM}_C^{\boldsymbol{e},\boldsymbol{m}}(n, L)$.

The $\mathbb{A}^1$–connectedness of $\mathcal{M}_C^{ss}(n, L)$ is shown in Section 2. Our proof is based on a result of Langton regarding the universal closedness of $\mathcal{M}_C^{ss}(n, L)$ [L]. To apply Langton for the semistable locus, one needs to have in place a direct $\mathbb{A}^1$-concordance between two K-points of $\mathcal{M}_C^{ss}(n, L)$ whose semistable locus is nonempty. This precludes the usage of the chain in [HY], as one can show that for $n > 2$, the intermediate $\mathbb{A}^1$s in their constructed concordance lie completely in the unstable locus. We avoid this by building a direct $\mathbb{A}^1$-concordance between any two vector bundles of the same rank and determinant. If we begin with two semistable vector bundles, then Langton applies non-trivially to give a direct $\mathbb{A}^1$-concordance which lies completely in the semistable locus. This also provides an alternate proof for a part of the main theorem of [HY].

In Section 3, we prove the $\mathbb{A}^1$–connectedness of $\mathcal{PM}_C^{\boldsymbol{e},\boldsymbol{m}}(n, L)$ using the concept of flag bundle stacks. In Section 4, $\mathbb{A}^1$-connectedness of the moduli stack $\mathcal{PM}_C^{\boldsymbol{\alpha}-ss,\boldsymbol{e},\boldsymbol{m}}(n, L)$ is shown for small and generic weights $\boldsymbol{\alpha}$ with $\gcd(n, \deg L) = 1$. In this case, $\boldsymbol{\alpha}$-semistability coincides with $\boldsymbol{\alpha}$-stability, the underlying vector bundles are stable, and the forgetful morphism to $\mathcal{M}_C^{s}(n, L)$ is an iterated flag bundle stack. The $\mathbb{A}^1$-connectedness then follows from Theorem 2.16 and Proposition 3.6.

2. $\mathbb{A}^1$–CONNECTEDNESS OF THE MODULI STACK OF SEMISTABLE VECTOR BUNDLES OVER A CURVE

2.1. $\mathbb{A}^1$-connected components of simplicial sheaves.

Definition 2.1 (Sheaf of $\mathbb{A}^1$-connected components)**.** Let S be a Noetherian base scheme of finite Krull dimension, and let Sm_S denote the category of smooth schemes of finite type over S, equipped with the Nisnevich topology. Let $\mathcal{H}_\bullet(S)$ denote the $\mathbb{A}^1$-homotopy category of spaces over S, which is the homotopy category obtained from the category of simplicial presheaves $\mathrm{sPre}(\mathrm{Sm}_S)$ by localizing with respect to $\mathbb{A}^1$-weak equivalences, as defined in [MV]. For a space $\mathcal{X} \in \mathcal{H}_\bullet(S)$, the **sheaf of $\mathbb{A}^1$-connected components**, denoted $\pi_0^{\mathbb{A}^1}(\mathcal{X})$, is the Nisnevich sheaf on Sm_S defined as the sheafification of the presheaf

$$U \longmapsto \mathrm{Hom}_{\mathcal{H}_\bullet(S)}(U, \mathcal{X}),$$

where $U \in \mathrm{Sm}_S$. Equivalently, if $\mathrm{L}_{\mathbb{A}^1} : \mathrm{sPre}(\mathrm{Sm}_S) \to \mathcal{H}_\bullet(S)$ denotes the $\mathbb{A}^1$-localization functor, then

$$\pi_0^{\mathbb{A}^1}(\mathcal{X}) = a_{\mathrm{Nis}}(U \mapsto \pi_0(\mathrm{L}_{\mathbb{A}^1}(\mathcal{X})(U)))\,,$$

where a_{Nis} denotes Nisnevich sheafification and π_0 denotes the set of path components of a simplicial set. A space $\mathcal{X}$ is called **$\mathbb{A}^1$-connected** if $\pi_0^{\mathbb{A}^1}(\mathcal{X}) \cong *$.

Definition 2.2 ([HY, Definition 2.6])**.** Let $\mathfrak{X}$ be a simplicial sheaf on Sm_k, the Grothendieck site of smooth finite-type schemes over a field k equipped with the Nisnevich topology.

(1) Let $U \in \mathrm{Sm}_k$. Two objects x and y in $\mathfrak{X}(U)$ are said to be *naively $\mathbb{A}^1$-homotopic* if there exists a map $f : \mathbb{A}^1 \times_k U \to \mathfrak{X}$ satisfying $f_0 = x$ and $f_1 = y$, where f_i is the composition $U \xrightarrow{i} \mathbb{A}^1 \times_k U \to \mathfrak{X}$.
(2) We shall denote by $S'(\mathfrak{X})$ the presheaf on Sm_k whose value on a k-scheme U is

$$U \mapsto \mathfrak{X}(U)/\!\sim,$$

where $\sim$ is the equivalence relation generated by naive $\mathbb{A}^1$-homotopy in (1). The Nisnevich sheafification of $S'(\mathfrak{X})$ is denoted by $S(\mathfrak{X})$.

Remark 2.3. By a result of Morel [M, Lemma 6.1.3], a space X over an infinite field k is $\mathbb{A}^1$-connected if and only if

$$\pi_0^{\mathbb{A}^1}(X)(\operatorname{Spec} F) = *$$

for every finitely generated separable field extension F of k. In addition, there exists a natural epimorphism $S(X) \to \pi_0^{\mathbb{A}^1}(X)$. Using this, one can show that if $S(X)(\operatorname{Spec} F) = *$ for every finitely generated separable field extension F of k, then X is $\mathbb{A}^1$–connected; this criterion is used in the proof of [HY, Theorem 1.1]. Moreover, since $S(X)$ is the Nisnevich sheafification of $S'(X)$ (see Definition 2.2), it follows from the properties of the Nisnevich topology that if $S'(X)(\operatorname{Spec} F) = *$ for every finitely generated separable field extension F of k, the same holds for its sheafification $S(X)$.

2.2. **Valuative criterion of properness for algebraic stacks.** Let $\mathfrak{X}$ be an algebraic stack of finite type over k. The valuative criterion of properness for $\mathfrak{X}$ states the following (see [ABBLT, § 3.1]): for every discrete valuation ring R with fraction field K and every 2-commutative diagram

$$\begin{array}{ccc} \operatorname{Spec} K & \longrightarrow & \mathfrak{X} \\ \downarrow & & \downarrow \\ \operatorname{Spec} R & \longrightarrow & \operatorname{Spec} k, \end{array} \tag{2.1}$$

there exists a field extension K'/K, a discrete valuation ring $R' \subseteq K'$ dominating R, and a morphism $\operatorname{Spec} R' \longrightarrow \mathfrak{X}$ making the following diagram 2-commutative:

$$\begin{array}{ccccc} \operatorname{Spec} K' & \longrightarrow & \operatorname{Spec} K & \longrightarrow & \mathfrak{X} \\ \downarrow & & \downarrow\ \exists & & \downarrow \\ \operatorname{Spec} R' & \longrightarrow & \operatorname{Spec} R & \longrightarrow & \operatorname{Spec} k. \end{array} \tag{2.2}$$

2.3. **Semistability of vector bundles.** The notion of semistability and stability of vector bundles over any irreducible smooth projective variety over a field is well-known; see [L, § 2] for the details. We recall it here for curves.

Definition 2.4. Let C denote an irreducible smooth projective curve over a field K, and let E be an algebraic vector bundle over C.

(i) The *degree* of E, denoted $\deg(E)$, is the degree of the divisor class associated to $\det(E)$.
(ii) The *slope* of E, denoted $\mu(E)$, is the rational number $\deg(E)/\mathrm{rank}(E)$.
(iii) E is *semistable* (respectively, *stable*) if for every proper sub-bundle $F \subsetneq E$ we have $\mu(F) \leq \mu(E)$ (respectively, $\mu(F) < \mu(E)$).

2.4. **Concordances.** The notion of $\mathbb{A}^1$–concordance between two vector bundles was introduced in [AKW]; we recall it below. Note that this notion coincides with naive $\mathbb{A}^1$-homotopy (Definition 2.2) in the context of moduli stacks of vector bundles over a curve.

Definition 2.5 ([AKW, Definition 5.2])**.** Let X be a scheme over a field K, and choose two distinct points $x_0, x_1 \in \mathbb{A}^1_K$. Two vector bundles $\mathcal{E}_0$ and $\mathcal{E}_1$ on X are said to be *directly* $\mathbb{A}^1$*-concordant* if there exists a vector bundle $\mathcal{E}$ over $X \times \mathbb{A}^1_K$ such that $\mathcal{E}_0 \xrightarrow{\simeq} \mathcal{E}|_{X\times\{x_0\}}$ and $\mathcal{E}_1 \xrightarrow{\simeq} \mathcal{E}|_{X\times\{x_1\}}$. They are $\mathbb{A}^1$*-concordant* if they are equivalent under the equivalence relation generated by direct $\mathbb{A}^1$-concordance.

Thus $\mathcal{E}_0$ and $\mathcal{E}_1$ are $\mathbb{A}^1$-concordant if there exists a finite sequence $\mathcal{E}_0 = \mathcal{F}_1, \mathcal{F}_2, \ldots, \mathcal{F}_r = \mathcal{E}_1$ such that each consecutive pair $(\mathcal{F}_i, \mathcal{F}_{i+1})$ is directly $\mathbb{A}^1$-concordant.

Definition 2.6. Let X be a scheme over a field K. Two semistable vector bundles $\mathcal{E}_0$ and $\mathcal{E}_1$ on X are *directly semistably* $\mathbb{A}^1$*-concordant* if there exists a family of semistable vector bundles $\mathcal{E}$ over $X \times \mathbb{A}^1_K$ (see Definition 2.8 (2)) such that $\mathcal{E}_0 \xrightarrow{\simeq} \mathcal{E}|_{X\times\{x_0\}}$ and $\mathcal{E}_1 \xrightarrow{\simeq} \mathcal{E}|_{X\times\{x_1\}}$ for two distinct points $x_0, x_1 \in \mathbb{A}^1_K$. They are *semistably* $\mathbb{A}^1$*-concordant* if they are equivalent under the equivalence relation generated by direct semistably $\mathbb{A}^1$-concordance.

The following standard result provides a large supply of direct $\mathbb{A}^1$-concordances; see also [HY, Proposition 2.4].

Proposition 2.7 ([La, Proposition 3.1])**.** *Let $V = \mathrm{Ext}^1(\mathcal{E}_1, \mathcal{E}_0)$, and let $p : C \times S \to C$ denote the projection for a k-scheme S. The functor sending S to the set of isomorphism classes of extensions of $p^*\mathcal{E}_1$ by $p^*\mathcal{E}_0$ on $C \times S$ is represented by the affine space $\mathbb{V}(V^\vee) \simeq \mathbb{A}^N_k$, where $N = \dim_K V$. In particular, for any extension $0 \to \mathcal{E}_0 \to \mathcal{E} \to \mathcal{E}_1 \to 0$, the bundle $\mathcal{E}$ is directly $\mathbb{A}^1$-concordant to $\mathcal{E}_0 \oplus \mathcal{E}_1$ via the family parametrized by the line $t \mapsto t \cdot [\mathcal{E}] \in \mathbb{A}^N_K$.*

Definition 2.8. Let C be an irreducible smooth projective curve over an algebraically closed field k, and fix a line bundle L on C. Let S be a k-scheme.

(1) A *family of vector bundles over C of rank n and determinant L parametrized by S* is a vector bundle $\mathcal{E}$ over $C \times S$ together with an isomorphism $\det(\mathcal{E}) \simeq p^*L$, where $C \times S \xrightarrow{p} C$ is the first projection.
(2) A family $\mathcal{E}$ as in (1) is a *family of semistable vector bundles* if for each geometric point $\mathrm{Spec}\, K \to S$ with K algebraically closed, the restriction $\mathcal{E}|_{C\times \mathrm{Spec}\, K}$ is semistable over $C_K := C \times \mathrm{Spec}\, K$.

Remark 2.9. For a k-scheme S, the S-valued points of $\mathcal{M}_C(n,L)$ and $\mathcal{M}_C^{ss}(n,L)$ are given by families of vector bundles and semistable vector bundles on C parametrized by S as in Definition 2.8, respectively.

One of the main inputs for proving the $\mathbb{A}^1$-connectedness of $\mathcal{M}_C^{ss}(n,L)$ is the following result, which guarantees the existence of semistable quotients. We first record a slope estimate.

Lemma 2.10. *There exists a constant a, depending only on the curve C, such that for any two vector bundles E and F over C,*

$$\mu_{\min}(E\otimes F)\geq \mu_{\min}(E)+\mu_{\min}(F)-a\,.$$

Proof. By [C, Proposition 2.2], $\mu_{\max}(E^\vee\otimes F^\vee)\leq \mu_{\max}(E^\vee)+\mu_{\max}(F^\vee)+a$ for some constant a depending only on C. The claim follows from the equality $\mu_{\max}(G^\vee)=-\mu_{\min}(G)$ for any vector bundle G on C. □

Proposition 2.11. *Let C be a smooth projective curve of genus $g\geq 2$ over an infinite field k with a rational point, and let E be a locally free sheaf of rank $n\geq 2$ and degree d on C. Then for any semistable locally free sheaf F of rank $n-1$ with $\det F\cong \det E(nm)$ and $m\gg 0$, there exists a surjection $E(m)\twoheadrightarrow F$.*

Proof. Set $\mathcal{H}:=\mathcal{H}om(E(m),F)$. By Lemma 2.10 there is a constant a depending only on C satisfying

$$\begin{aligned}\mu_{\min}(\mathcal{H})=\mu_{\min}(E^\vee\otimes F(-m)) &\geq \mu_{\min}(E^\vee)+\mu_{\min}(F(-m))-a\\ &=\mu_{\min}(F(-m))+c, \qquad c:=\mu_{\min}(E^\vee)-a\\ &=\mu(F)-m+c\\ &=\frac{d+mn}{n-1}-m+c\\ &=\frac{m}{n-1}+\Big(\frac{d}{n-1}+c\Big),\end{aligned}$$

where $d=\deg(E)$. Hence $\mu_{\min}(\mathcal{H})\to+\infty$ as $m\to\infty$. In particular, by Lemma 2.10,

$$\mu_{\max}(\mathcal{H}^\vee(x)\otimes\omega_C)=-\mu_{\min}(\mathcal{H}\otimes\omega_C(x))<0$$

for all $x\in C$ and $m\gg 0$. Since a nonzero global section of a bundle of negative maximal slope would yield a degree-zero subbundle, we get $H^0(C,\mathcal{H}^\vee(x)\otimes\omega_C)=0$, hence $H^1(C,\mathcal{H}(-x))=0$ for all $x\in C$ by Serre duality. Thus $\mathcal{H}$ is globally generated.

Let $V:=H^0(C,\mathcal{H})$; by Riemann–Roch, $\dim V\to\infty$ as $m\to\infty$. Consider the incidence locus

$$Z=\{(\phi,x)\in\mathbb{A}(V)\times C\mid \phi_x \text{ not surjective}\}.$$

For each $x\in C$, the evaluation map $\mathrm{ev}_x: V\twoheadrightarrow \mathrm{Hom}(k^n,k^{n-1})$ is a surjective linear map, hence smooth. The locus Y in $\mathrm{Hom}(k^n,k^{n-1})$ of non-surjective maps is the determinantal variety of maps of rank $\leq n-2$, which has codimension at least 2. The fiber Z_x of the projection $p: Z\longrightarrow C$ is $\mathrm{ev}_x^{-1}(Y)$; by smoothness of ev_x, it also has codimension at least 2 in V, so $\dim Z_x\leq \dim V-2$. Since p is surjective, by [Har, II § 3, Ex. 3.22]

$$\dim Z\leq \dim Z_x+\dim C=\dim\mathbb{A}(V)-1<\dim\mathbb{A}(V).$$

The image of $Z \longrightarrow \mathbb{A}(V)$ is therefore a proper closed subvariety. Since k is infinite, there exists $\phi \in V(k)$ in the open complement, giving the required surjection $E(m) \twoheadrightarrow F$. □

Remark 2.12. The existence of a semistable quotient of this type is likely known to experts; see the remark following [PR, Proposition 6.1]. We include a proof here as our intended application requires the most general assumptions on the base field.

Proposition 2.13. *Let F be a field extension of k. Let $\mathcal{E}_0$ and $\mathcal{E}_1$ be two vector bundles of rank n and determinant p^*L over the curve $C_F = C \times_k \operatorname{Spec} F$, where $p : C_F \to C$ is the projection and $L \in \operatorname{Pic}(C)$. Then $\mathcal{E}_0$ and $\mathcal{E}_1$ are directly $\mathbb{A}^1$-concordant.*

Proof. The curve C_F is an irreducible smooth projective curve over F; smoothness and projectivity are stable under base change, while irreducibility follows from [EGA, IV_2, Proposition 4.5.9]. Choose any semistable vector bundle $\mathcal{F}$ of rank $n-1$ and determinant $p^*L(mn)$ on C_F. By applying Proposition 2.11 to both $\mathcal{E}_0$ and $\mathcal{E}_1$, we may choose $m \gg 0$ so that there exist surjections $\mathcal{E}_i(m) \twoheadrightarrow \mathcal{F}$, $i = 0, 1$, with kernels K_i satisfying $K_i \simeq \mathcal{O}_{C_F}$ (by comparison of determinants). Twisting by $\mathcal{O}(-m)$ gives extensions

$$0 \longrightarrow \mathcal{O}_{C_F}(-m) \longrightarrow \mathcal{E}_i \longrightarrow \mathcal{F}(-m) \longrightarrow 0\,, \quad i = 0, 1. \tag{2.3}$$

These give rise to two extension classes $[\mathcal{E}_i] \in V := \operatorname{Ext}^1\big(\mathcal{F}(-m), \mathcal{O}_{C_F}(-m)\big)$. The straight line $t \mapsto (1-t)[\mathcal{E}_0] + t[\mathcal{E}_1]$ in V joins the two classes, and by Proposition 2.7 this yields a direct $\mathbb{A}^1$-concordance between $\mathcal{E}_0$ and $\mathcal{E}_1$. □

Remark 2.14. This strengthens [HY, Theorem 2.5(b)]: for vector bundles $\mathcal{E}$ and $\mathcal{F}$ of rank n over C, they are directly $\mathbb{A}^1$-concordant if and only if $\det(\mathcal{E}) \simeq \det(\mathcal{F})$. One direction is Proposition 2.13; for the converse, if $\mathcal{E}$ and $\mathcal{F}$ are directly concordant, there is an induced morphism $\mathbb{A}^1 \to \operatorname{Pic}^d(C)$, which must be constant as $\operatorname{Pic}^d(C)$ is an abelian variety. Thus $\det(\mathcal{E}) \simeq \det(\mathcal{F})$.

Proposition 2.15. *With the same setup as in Proposition 2.13, assume moreover that $\mathcal{E}_0$ and $\mathcal{E}_1$ are semistable. Then there exists a direct semistable $\mathbb{A}^1$-concordance between $\mathcal{E}_0$ and $\mathcal{E}_1$; see Definition 2.6.*

Proof. Proposition 2.13 yields a vector bundle $\mathcal{E}$ on $C_F \times_F \mathbb{A}^1_F$ and distinct points $x_0, x_1 \in \mathbb{A}^1_F$ satisfying Definition 2.5. Consider the Cartesian diagram with π and π' the first projection maps:

$$\begin{array}{ccccc} C \times_k \mathbb{A}^1_F & \xrightarrow{\simeq} & C_F \times_F \mathbb{A}^1_F & \xrightarrow{\pi'} & C_F \\ \downarrow{\scriptstyle p'} & & & & \downarrow{\scriptstyle p} \\ C \times_k \mathbb{A}^1_k & & \xrightarrow{\qquad\pi\qquad} & & C. \end{array} \tag{2.4}$$

From the construction of $\mathcal{E}$ in Proposition 2.13, it follows that

$$\det(\mathcal{E}) \;\simeq\; (\pi')^* \det(\mathcal{E}_i) \;\simeq\; (\pi')^*(p^*L) \;=\; (p \circ \pi')^* L = (\pi \circ p')^* L, \quad i = 0, 1. \tag{2.5}$$

Via the identification $C \times_k \mathbb{A}^1_F \simeq C_F \times_F \mathbb{A}^1_F$, the map $\pi \circ p'$ corresponds to the first projection $p_1 : C \times_k \mathbb{A}^1_F \longrightarrow C$. Regarding $\mathcal{E}$ as a vector bundle on $C \times_k \mathbb{A}^1_F$, equation (2.5)

gives $\det(\mathcal{E}) \simeq p_1^* L$. By Definition 2.8, $\mathcal{E}$ is a family of vector bundles on C of rank n and determinant L parametrized by $\mathbb{A}^1_F$, which by the 2-Yoneda lemma gives a morphism

$$\varphi \;:\; \mathbb{A}^1_F \;\longrightarrow\; \mathcal{M}_C(n, L)\,. \tag{2.6}$$

By [ABBLT, Proposition 2.11], the semistable locus $U := \{p \in \mathbb{A}^1_F \mid \mathcal{E}|_{C\times\{p\}} \text{ semistable}\}$ is an open subscheme of $\mathbb{A}^1_F$, and $\varphi|_U$ factors through $\mathcal{M}^{ss}_C(n, L)$:

$$\begin{array}{ccc} \mathbb{A}^1_F & \xrightarrow{\;\varphi\;} & \mathcal{M}_C(n, L) \\ \uparrow & & \uparrow \\ U & \xrightarrow{\;\varphi|_U\;} & \mathcal{M}^{ss}_C(n, L)\,. \end{array} \tag{2.7}$$

The stack $\mathcal{M}^{ss}_C(n, L)$ is universally closed over k by a result of Langton [L, Proposition 6] (see also [ABBLT, Theorem 3.7]):

Let R be a discrete valuation ring with fraction field K. For any semistable vector bundle E of fixed rank and determinant on C_K, there exists a vector bundle $\mathcal{E}$ on C_R with $\mathcal{E}_K \simeq E$ and semistable special fiber.

In the language of stacks, this translates via the 2-Yoneda lemma to:

$$\begin{array}{ccc} \operatorname{Spec} K & \longrightarrow & \mathcal{M}^{ss}_C(n, L) \\ \downarrow & \overset{\exists}{\nearrow} & \downarrow \\ \operatorname{Spec} R & \longrightarrow & \operatorname{Spec} k. \end{array} \tag{2.8}$$

In other words, the valuative criterion for $\mathcal{M}^{ss}_C(n, L)$ does not require further field extensions: it suffices to take $K' = K$ and $R' = R$ in diagram (2.2) (see [ABBLT, § 3.1]). Hence $\varphi|_U : U \longrightarrow \mathcal{M}^{ss}_C(n, L)$ extends to a morphism

$$\widetilde{\varphi} \;:\; \mathbb{A}^1_F \;\longrightarrow\; \mathcal{M}^{ss}_C(n, L),$$

which by the 2-Yoneda lemma corresponds to a family $\widetilde{\mathcal{E}}$ of semistable vector bundles over C of rank n and determinant L parametrized by $\mathbb{A}^1_F$. Since $\mathcal{E}_0$ and $\mathcal{E}_1$ are semistable, the points x_0 and x_1 lie in U and are unchanged by $\widetilde{\varphi}$. Therefore

$$\widetilde{\mathcal{E}}|_{C_F\times\{x_j\}} \;\simeq\; \mathcal{E}_j, \quad j = 0, 1,$$

showing that $\mathcal{E}_0$ and $\mathcal{E}_1$ are directly semistably $\mathbb{A}^1$-concordant. □

Theorem 2.16. *The moduli stack $\mathcal{M}^{ss}_C(n, L)$ of semistable vector bundles of rank n and determinant L over an irreducible smooth projective curve C is $\mathbb{A}^1$–connected. In particular, if n and $\deg(L)$ are coprime, the moduli stack of stable vector bundles $\mathcal{M}^{s}_C(n, L)$ is $\mathbb{A}^1$–connected.*

Proof. This follows immediately from Remark 2.3 and Proposition 2.15. □

3. $\mathbb{A}^1$–CONNECTEDNESS OF THE MODULI STACK OF QUASI-PARABOLIC VECTOR BUNDLES OVER A CURVE

The notion of parabolic vector bundles over a curve was introduced by Mehta and Seshadri in [MS]. As before, C denotes an irreducible smooth projective curve over an algebraically closed field k. Fix a finite subset $D \subset C$ of distinct closed points, called *parabolic points*, and the following data:

- a positive integer n,
- for each $p \in D$, a positive integer $e(p)$ and a tuple of positive integers $\underline{m}(p) := (m_{p,1}, \ldots, m_{p,e(p)})$ satisfying $\sum_{i=1}^{e(p)} m_{p,i} = n$.

Set $\boldsymbol{e} := \{e(p) \mid p \in D\}$ and $\boldsymbol{m} := \{\underline{m}(p) \mid p \in D\}$.

Definition 3.1. The tuple $(\boldsymbol{e}, \boldsymbol{m})$ is called a *quasi-parabolic data of rank n along D*. The collection $\boldsymbol{e}$ is a system of *lengths* along D, and $\boldsymbol{m}$ is a system of *multiplicities* along D.

Definition 3.2. Fix a quasi-parabolic data $(\boldsymbol{e}, \boldsymbol{m})$. A *quasi-parabolic vector bundle of rank n* on C is a rank n vector bundle E on C together with, for each $p \in D$, a flag of type $\underline{m}(p)$ on the fiber E_p:

$$E_p = E_{p,1} \supsetneq E_{p,2} \supsetneq \cdots \supsetneq E_{p,e(p)} \supsetneq E_{p,e(p)+1} = 0,$$

satisfying $\dim(E_{p,i}/E_{p,i+1}) = m_{p,i}$ for all $1 \leq i \leq e(p)$.

3.1. **Moduli stack of quasi-parabolic vector bundles.** For a vector bundle $\mathcal{F}$ of rank n on a k-scheme S and a tuple $\underline{m} = (m_1, \ldots, m_e)$ with $\sum m_i = n$, let $\mathrm{Flag}_{\underline{m}}(\mathcal{F}) \longrightarrow S$ denote the associated flag bundle of type $\underline{m}$. By its universal property, a section of this flag bundle corresponds to a chain of sub-bundles of $\mathcal{F}$ of type $\underline{m}$.

Definition 3.3 ([BY, Definition 3.1])**.** A *family of quasi-parabolic vector bundles of type $(\boldsymbol{e}, \boldsymbol{m})$ on C parametrized by S* is a vector bundle $\mathcal{F}$ on $C \times S$ together with, for each $p \in D$, a section s_p of $\mathrm{Flag}_{\underline{m}(p)}\left(\mathcal{F}|_{\{p\}\times S}\right) \longrightarrow S$.

The moduli stack of quasi-parabolic vector bundles on C of rank n, determinant $L \in \mathrm{Pic}(C)$ and quasi-parabolic data $(\boldsymbol{e}, \boldsymbol{m})$ is denoted $\mathcal{PM}_C^{\boldsymbol{e},\boldsymbol{m}}(n, L)$. Its objects over S are families as in Definition 3.3 with $\det(\mathcal{F}) \simeq p^*L$.

Definition 3.4. Let $\mathfrak{X}$ be an algebraic stack over k, and let $\mathcal{E}$ be a vector bundle of rank n over $\mathfrak{X}$. Fix a tuple $\underline{m} = (m_1, \ldots, m_e)$ with $\sum m_i = n$. The *flag bundle stack of type $\underline{m}$ associated to $\mathcal{E}$*, denoted $\mathrm{Flag}_{\underline{m}}(\mathcal{E})$, is the algebraic stack whose T-points are pairs (u, F) with $u : T \to \mathfrak{X}$ a morphism and F a flag of type $\underline{m}$ on $u^*\mathcal{E}$.

Remark 3.5. If $\mathcal{E}$ trivializes over a Zariski cover $V \to \mathfrak{X}$, then $\mathrm{Flag}_{\underline{m}}(\mathcal{E}) \times_{\mathfrak{X}} V \simeq \mathrm{Flag}_{\underline{m}}(k^n) \times V$. Thus $\mathrm{Flag}_{\underline{m}}(\mathcal{E})$ is Zariski-locally trivial with fiber the flag variety $\mathrm{Flag}_{\underline{m}}(k^n)$.

Proposition 3.6. *Let $\mathfrak{X}$ be an algebraic stack over an algebraically closed field k, let F be a finitely generated field extension of k, and let $f : \mathfrak{Y} \longrightarrow \mathfrak{X}$ be a Zariski-locally trivial flag bundle stack with fibers isomorphic to a flag variety Z. If $S'(\mathfrak{X})(\mathrm{Spec}\, F) = 0$, then $S'(\mathfrak{Y})(\mathrm{Spec}\, F) = 0$.*

Proof. Given two F-points $x, y \in \mathfrak{Y}$, we show they can be joined by a finite sequence of maps $\mathbb{A}^1_F \longrightarrow \mathfrak{Y}$.

If $f(x) = f(y)$ in $\mathfrak{X}$, then x and y lie in the same fiber of f, which is a flag variety. Since flag varieties are rational and hence $\mathbb{A}^1$-chain connected, x and y can be joined.

If $f(x) \neq f(y)$, by hypothesis there is a finite sequence of maps $\mathbb{A}^1_F \to \mathfrak{X}$ joining $f(x)$ to $f(y)$; without loss of generality assume $f(x)$ and $f(y)$ are directly joined by a morphism $g : \mathbb{A}^1_F \longrightarrow \mathfrak{X}$ with $g(0) = f(x)$ and $g(1) = f(y)$. Since f is a representable Zariski-locally trivial flag bundle stack, the pullback $g^*\mathfrak{Y} \longrightarrow \mathbb{A}^1_F$ is the trivial flag bundle $\mathbb{A}^1_F \times_k Z$. Let x' and y' denote the points of $\mathbb{A}^1_F \times_k Z$ corresponding to $(0, x)$ and $(1, y)$ respectively. Since $\mathbb{A}^1_F$ and Z are both $\mathbb{A}^1$-chain connected, so is their product, and x' and y' can be joined by a sequence of maps $\mathbb{A}^1_F \to \mathbb{A}^1_F \times_k Z \simeq g^*\mathfrak{Y}$. Composing with $g^*\mathfrak{Y} \to \mathfrak{Y}$ completes the proof. □

Theorem 3.7. *The moduli stack of quasi-parabolic vector bundles $\mathcal{PM}_C^{\boldsymbol{e},\boldsymbol{m}}(n, L)$ is $\mathbb{A}^1$–connected.*

Proof. The forgetful map

$$\mathcal{PM}_C^{\boldsymbol{e},\boldsymbol{m}}(n, L) \longrightarrow \mathcal{M}_C(n, L) \tag{3.1}$$

is an iterated Zariski-locally trivial flag bundle stack. For $D = \{p\}$ a single point, letting $\mathcal{E}$ be the universal bundle over $\mathcal{M}_C(n, L) \times C$ and $\iota_p : \mathcal{M}_C(n, L) \hookrightarrow \mathcal{M}_C(n, L) \times C$ the inclusion at p, the S-points of $\mathcal{PM}_C^{\boldsymbol{e},\boldsymbol{m}}(n, L)$ and $\mathrm{Flag}_{\underline{m}}(\iota_p^*\mathcal{E})$ coincide for any k-scheme S, giving an isomorphism

$$\mathcal{PM}_C^{\boldsymbol{e},\boldsymbol{m}}(n, L) \xrightarrow{\simeq} \mathrm{Flag}_{\underline{m}}(\iota_p^*\mathcal{E})\,.$$

For $D = \{p_1, \ldots, p_r\}$, the stack $\mathcal{PM}_C^{\boldsymbol{e},\boldsymbol{m}}(n, L)$ is the fiber product over $\mathcal{M}_C(n, L)$ of the stacks for each p_i, confirming that (3.1) is an iterated flag bundle stack.

By [HY, Theorem 1.1], $S'(\mathcal{M}_C(n, L))(\mathrm{Spec}\, F) = 0$ for every finitely generated field extension F/k. Applying Proposition 3.6 iteratively gives $S'(\mathcal{PM}_C^{\boldsymbol{e},\boldsymbol{m}}(n, L))(\mathrm{Spec}\, F) = 0$, and the $\mathbb{A}^1$-connectedness follows from Remark 2.3. □

4. $\mathbb{A}^1$–CONNECTEDNESS OF THE MODULI STACK OF PARABOLIC SEMISTABLE BUNDLES OVER A CURVE

We now consider the open substacks of $\mathcal{PM}_C^{\boldsymbol{e},\boldsymbol{m}}(n, L)$ consisting of $\boldsymbol{\alpha}$-(semi)stable parabolic bundles. Fix a quasi-parabolic data $(\boldsymbol{e}, \boldsymbol{m})$ of rank n along D, and a system of weights

$$\boldsymbol{\alpha} := \{(\alpha_{p,1} < \alpha_{p,2} < \cdots < \alpha_{p,e(p)}) \mid p \in D\}, \tag{4.1}$$

where $\alpha_{p,i} \in [0, 1)$ for all p and i.

Definition 4.1. Let E be a quasi-parabolic vector bundle of rank n on C associated to $(\boldsymbol{e}, \boldsymbol{m})$ (see Definition 3.2).

(1) The $\boldsymbol{\alpha}$-*degree* of E is

$$\deg_{\boldsymbol{\alpha}}(E) := \deg(E) + \sum_{p \in D} \sum_{i=1}^{e(p)} m_{p,i}\, \alpha_{p,i}.$$

(2) The $\boldsymbol{\alpha}$-*slope* of E is $\mu_{\boldsymbol{\alpha}}(E) := \deg_{\boldsymbol{\alpha}}(E)/n$.

The notion of $\boldsymbol{\alpha}$-semistability and $\boldsymbol{\alpha}$-stability for a quasi-parabolic vector bundle is well-known; see [MS, Definition 1.13]. The quasi-parabolic moduli stack $\mathcal{PM}^{\boldsymbol{e},\boldsymbol{m}}(n,L)$ contains an open substack of $\boldsymbol{\alpha}$-semistable bundles, denoted $\mathcal{PM}_C^{\boldsymbol{\alpha}\text{-ss},\boldsymbol{e},\boldsymbol{m}}(n,L)$.

Definition 4.2. A system of weights $\boldsymbol{\alpha}$ is said to be *generic* if a quasi-parabolic vector bundle is $\boldsymbol{\alpha}$-semistable if and only if it is $\boldsymbol{\alpha}$-stable. We shall call a system of weights $\boldsymbol{\alpha}$ to be *small* if it satisfies the numerical condition of [BY, Proposition 5.3]. We refer to [BY, § 2 and § 5] for more details.

The notion of small and generic weights will be useful for our next result.

Theorem 4.3. *For $\boldsymbol{\alpha}$ small and generic and $\gcd(n,\deg L)=1$, the moduli stack $\mathcal{PM}_C^{\boldsymbol{\alpha}\text{-ss},\boldsymbol{e},\boldsymbol{m}}(n,L)$ is $\mathbb{A}^1$–connected.*

Proof. Under these assumptions, $\boldsymbol{\alpha}$-semistability coincides with $\boldsymbol{\alpha}$-stability and every semistable bundle of rank n and degree $\deg L$ is stable, so $\mathcal{PM}_C^{\boldsymbol{\alpha}\text{-ss},\boldsymbol{e},\boldsymbol{m}}(n,L)=\mathcal{PM}_C^{\boldsymbol{\alpha}\text{-s},\boldsymbol{e},\boldsymbol{m}}(n,L)$ and $\mathcal{M}_C^{\mathrm{ss}}(n,L)=\mathcal{M}_C^{\mathrm{s}}(n,L)$. By [BY, Proposition 5.3], the forgetful morphism

$$\mathcal{PM}_C^{\boldsymbol{\alpha}\text{-s},\boldsymbol{e},\boldsymbol{m}}(n,L)\longrightarrow\mathcal{M}_C^{\mathrm{s}}(n,L)$$

is an iterated flag bundle stack. Applying Proposition 3.6 together with Theorem 2.16 gives $S'(\mathcal{PM}_C^{\boldsymbol{\alpha}\text{-s},\boldsymbol{e},\boldsymbol{m}}(n,L))(F)=*$, establishing $\mathbb{A}^1$-connectedness using Remark 2.3. □

Remark 4.4. In the above proof we do not need to invoke Langton to repair paths: by By [BY, Proposition 5.3], the paths constructed in Proposition 3.6 consist entirely of $\boldsymbol{\alpha}$-semistable points.

Remark 4.5. The assumptions of Theorem 4.3 may appear restrictive. For general $\boldsymbol{\alpha}$, the methods of this paper do not yield a direct $\mathbb{A}^1$-concordance between two K-points of $\mathcal{PM}_C^{\boldsymbol{\alpha}\text{-ss},\boldsymbol{e},\boldsymbol{m}}(n,L)$: the concordance in Proposition 3.6 consists of multiple $\mathbb{A}^1$-paths, and it is not clear that each path has a nonempty $\boldsymbol{\alpha}$-semistable locus, which would be needed to apply Langton.

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